

ASSIGNMENT 1 — COMPLEX PROBLEM-SOLVING / MINI PROJECT

Design, Simulation and Material Transport Integration for an Automated Flexible Manufacturing Cell

*Technical Design Document · Machine specification, feed drive and feedback selection,
CNC part programming with collision verification, and AGV throughput analysis*

Particulars	Details
Name of the student	
Enrolment / Roll number · Class	
Subject	Automation in Manufacturing · PBL Assignment 1
Target COs / SDGs	CO-2, CO-3, CO-4 · SDG 9, SDG 12
Date of submission	

1. Scope

A precision automotive component supplier is converting a manual job shop into an Automated Flexible Manufacturing Cell. The cell must hold a **repeatable positioning accuracy of ± 0.005 mm** under continuous operation. This document contains the three deliverables required: the **machine specification sheet**, the **feed drive and feedback sensor selection justification**, and the **AGV throughput calculations**, together with the verified part programs and the cell layout diagram.

2. Machine Specification Sheet

Two machine tools cover the part family of rotationally symmetric shafts and prismatic plates.

Parameter	TC-01 — CNC turning centre	MC-01 — 3-axis CNC VMC
Controlled axes	2 (X, Z), C-axis ready	3 (X, Y, Z), 4th axis ready
Travel	X 210 mm · Z 560 mm	X 760 · Y 460 · Z 510 mm
Capacity	Ø340 mm swing, Ø65 mm bar	900 × 460 mm table
Spindle	A2-6 nose, 15 kW, 4 500 rev/min	BT-40, 11 kW, 12 000 rev/min
Spindle bearings	Hybrid ceramic angular contact, DBB set	Hybrid ceramic angular contact, 4 × DBB set
Feed drive	Ø32 × 10 mm C3 ground ball screw, double-nut preload, direct-coupled AC servo	Same, all three axes

Parameter	TC-01 — CNC turning centre	MC-01 — 3-axis CNC VMC
Guideways	Roller LM guides (X) + hardened and ground box ways (Z)	Roller LM guides, preload class C1, all axes
Position feedback	Absolute linear glass scale, both axes (full closed loop)	Absolute linear glass scale, all axes (full closed loop)
Rapid traverse	24 / 30 m/min	36 / 36 / 30 m/min
Positioning accuracy	± 0.005 mm	± 0.005 mm
Repeatability	± 0.002 mm	± 0.002 mm
Tool changer	12-station servo turret, 0.15 s index	24-pocket ATC, 1.3 s tool-to-tool
Control	Fanuc Oi-TF	Fanuc Oi-MF
Automation interface	Robot I/O, auto door, parts catcher	Robot I/O, auto door, chip conveyor

3. Feed Drive and Feedback Sensor Selection Justification

± 0.005 mm is a **system** requirement. Each contributor to positioning error is identified, a component selected so that its share is small, and the residuals combined into an error budget.

3.1 Recirculating ball screw

Parameter	Selection	Justification
Nominal diameter	Ø 32 mm	Adequate buckling and critical-speed margin over the 760 mm X travel while keeping rotary inertia low enough for a 36 m/min rapid.
Lead (pitch)	10 mm	36 m/min needs 3 600 rev/min; $d \cdot n = 32 \times 3\,600 = 1.15 \times 10^5$, inside the 1.6×10^5 limit for a ground screw with forced lubrication. A 5 mm lead would halve the available speed.
Accuracy grade	JIS C3 (ISO 5), ± 8 µm / 300 mm	Normal grade for single-digit micrometre work. C5 leaves too little budget; C1/C2 is laboratory grade and not cost-justified.
Preloading method	Double-nut (or offset-lead) preload at 7 % of C_a	Eliminates axial backlash, which otherwise dominates reversal error. 7 % is the standard compromise — enough to remove lost motion, low enough to avoid excess heat and preload collapse.
Mounting	Fixed-fixed, axially pre-tensioned angular contact support bearings	Roughly doubles axial stiffness and absorbs the thermal growth of the screw instead of displacing the table.
Drive	Direct bellows coupling to the AC servo	Removes the backlash and torsional wind-up a belt or gear reduction would add.

Resolution check. With a 10 mm lead and a 2^{20} count/rev encoder, one count = $10 / 1\,048\,576 = \mathbf{0.0000095\text{ mm}}$. Command resolution is therefore not the limit; the mechanical and metrological chain is, which is why the loop is closed on a linear scale.

3.2 Linear guideways — LM guides versus hardened and ground box ways

Criterion	Recirculating LM (rolling) guideways	Hardened and ground box (sliding) ways
Friction	$\mu \approx 0.002\text{--}0.005$, static $\approx$ dynamic	$\mu \approx 0.05\text{--}0.15$, static $>$ dynamic
Stick-slip at low feed	Virtually absent — decisive for micrometre positioning	Present; jerk at reversal and following error at creep feeds
Reversal error	$\approx 1\text{ }\mu\text{m}$ with preload	3–10 μm
Damping	Low	High — the oil film over large sliding areas damps chatter
Stiffness	High, raised by preload class (C0 $\rightarrow$ C1 $\rightarrow$ C2)	Very high over the large contact area
Rapid traverse	Over 100 m/min	Limited by friction and heat, typically $< 20\text{ m/min}$
Maintenance	Sealed, grease- or oil-lubricated, replaceable blocks	Way lubrication, wipers, periodic re-scraping
Cost	Moderate, bought in	High labour content (hand scraping)

Selection — roller-type LM guideways, accuracy class P, preload class C1 on all three MC-01 axes and the TC-01 X axis. Roller blocks are chosen over ball blocks because line contact gives roughly twice the stiffness in the same envelope, and the absence of stick-slip is what makes $\pm 0.005\text{ mm}$ reachable at finishing feeds. **The TC-01 Z axis keeps hardened and ground box ways**, because it carries the heavy roughing loads where the damping of a sliding way suppresses chatter; its accuracy is secured by closing the loop on a linear scale.

3.3 High-speed spindle bearings

Selection: hybrid ceramic angular contact ball bearings — silicon nitride (Si_3N_4) balls in steel rings — as a four-bearing back-to-back set, 15° contact angle at the front and 25° at the rear, light-to-medium preload, oil-air lubrication.

Property of Si_3N_4 balls	Relative to steel	Consequence
Density	$\approx 40\%$ (3.2 vs 7.8 g/cm ³)	Centrifugal force and gyroscopic moment fall $\sim 60\%$, so the contact angle stays stable and the speed limit rises
Modulus of elasticity	$\approx 1.5 \times$ (310 vs 210 GPa)	Smaller contact deformation, so higher radial and axial stiffness — accuracy held while cutting
Thermal expansion	$\approx 25\%$	Less preload growth as the spindle warms: less heat, far less risk of preload collapse
Hardness, inertness, non-conductive	Harder, chemically inert	No micro-welding with the rings, better life under marginal lubrication, immune to VFD shaft-current pitting

Speed check. A 70 mm bore front pair has $d_m \approx 92.5\text{ mm}$. At 12 000 rev/min, $d_m \cdot n = \mathbf{1.11 \times 10^6\text{ mm}\cdot\text{rev/min}}$. Grease would be marginal here, so **oil-air lubrication is specified**, under which hybrid ceramic angular contact bearings are rated to about 1.6×10^6 — a comfortable margin.

3.4 Position feedback — optical rotary encoder versus linear glass scale

This decides **which errors the control can see and correct**.

Aspect	Rotary encoder on the screw (semi-closed loop)	Linear glass scale on the axis (full closed loop)
What it measures	The angle of the ball screw ; table position inferred from the lead	The actual position of the table , directly
Ball screw pitch error	Not seen; only reducible by stored pitch-error compensation	Seen and corrected automatically
Thermal growth of the screw	Not seen at all — 2 °C over 500 mm gives $\approx 12\ \mu\text{m}$ of uncorrected error	Seen and corrected — the scale measures the table, not the screw
Backlash, axial play in nut and supports	Not seen	Seen and corrected
Resolution	0.0001 mm equivalent	0.0001 mm (0.1 μm), grades to 0.00005 mm
System accuracy achievable	± 0.010 to ± 0.015 mm	± 0.003 to ± 0.005 mm
Loop stability	Easier — the structure is outside the loop	Harder — structural compliance is inside the loop, so the drive needs careful tuning
Relative cost	Low	Higher (scale, reader head, mounting, protection)

Selection — absolute linear glass scales, 0.1 μm resolution, $\pm 3\ \mu\text{m}/\text{m}$ accuracy grade, sealed and dry-air purged, on every linear axis, with the motor encoder retained for velocity feedback and commutation. A semi-closed loop cannot meet ± 0.005 mm in a continuously running cell: thermal growth of the ball screw alone exceeds the whole error budget within the first hour.

3.5 Error budget

Contributions are combined as the root sum of squares, evaluated at a 500 mm axis position.

#	Error source	Raw magnitude	Mitigation	Residual, μm
1	Ball screw lead error (C3)	$\pm 8\ \mu\text{m} / 300\ \text{mm}$	Laser-measured stored pitch-error compensation	± 2.0
2	Thermal growth of the screw	$\approx 12\ \mu\text{m}$ at $\Delta T = 2\ ^\circ\text{C}$	Loop closed on the linear scale; fixed-fixed pre-tensioned mounting	± 1.0
3	Linear scale accuracy grade	$\pm 3\ \mu\text{m} / \text{m}$	Graduation accuracy over 500 mm	± 1.5
4	Reversal error (backlash + stick-slip)	3–10 μm if unaddressed	Double-nut preload + preloaded roller LM guides + direct coupling	± 1.0
5	Structural and Abbe offset error	2–4 μm	Scale mounted close to the work plane	± 1.5
6	Resolution / quantisation	0.1 μm	Scale interpolation	± 0.05
	Root sum of squares		$\sqrt{(2.0^2 + 1.0^2 + 1.5^2 + 1.0^2 + 1.5^2 + 0.05^2)}$	± 3.24

Result — predicted positioning accuracy $\pm 3.24 \mu\text{m}$ against the $\pm 5 \mu\text{m}$ requirement, a margin of about 35 % that absorbs foundation settlement, ambient drift across a shift and preload wear. Verify on site by laser interferometry to ISO 230-2 and re-measure the compensation tables annually.

4. CNC Part Programming and Collision Verification

	O0001 — turning	O0002 — milling
Component	Stepped transmission shaft, mild steel	Cover plate, aluminium
Stock	Ø50 × 100 mm bar	160 × 100 × 20 mm plate
Features programmed	Facing (G94) · two-step stock removal (G90) · 2 × 45° chamfer · 4 × 2 mm relief groove · M30 × 1.5 thread (G92) · G42 nose radius compensation	Ø63 face milling · 60 × 40 × 6 rectangular pocket · Ø40 × 6 circular pocket · 4 × Ø8 holes (G83 peck) · G41 cutter radius compensation · subprograms O1001 and O1002
Full listing	Appendix A	Appendix B

Both programs were verified with a G-code interpreter and material-removal simulator (simulation/toolpath_simulator.py). It expands the subprogram calls, the lathe one-shot cycles and the milling canned cycles, applies G41/G42 compensation, and removes material from a model of the stock — a radius profile $r(z)$ for turning and a height map $z(x, y)$ for milling. Every rapid traverse is tested against the stock **as it stands at that moment**.

Check	O0001 turning	O0002 milling
Elementary moves interpreted	83	381
Rapid traverse into remaining material	none	none
Travel outside the machine envelope	none	none
Cutting move with spindle stopped or no feed set	none	none
Tool change with compensation still active	none	none
Dimensional checks on the simulated part	profile matches the drawing	11 of 11 passed
Verification result	PASS — collision free	PASS — collision free

Validity of the PASS result — the simulator also runs two **deliberately faulty** programs as a negative control: one rapids straight into the uncut bar, the other cuts with the spindle stopped, with no feed rate, and rapid-plunges into solid stock. The checker detects 1 and 3 faults respectively, so the PASS results are meaningful. Full output in Appendix C.

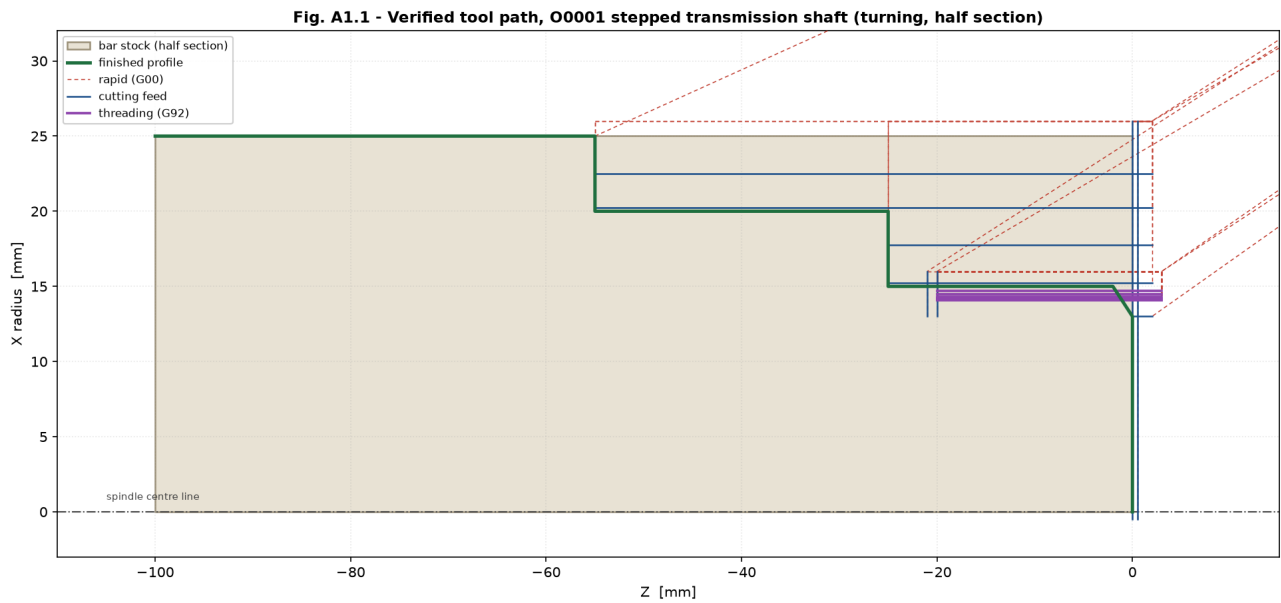


Fig. 1 — Verified tool path, O0001 (half section). Broken red = rapid, solid blue = cutting feed, green = finished profile.

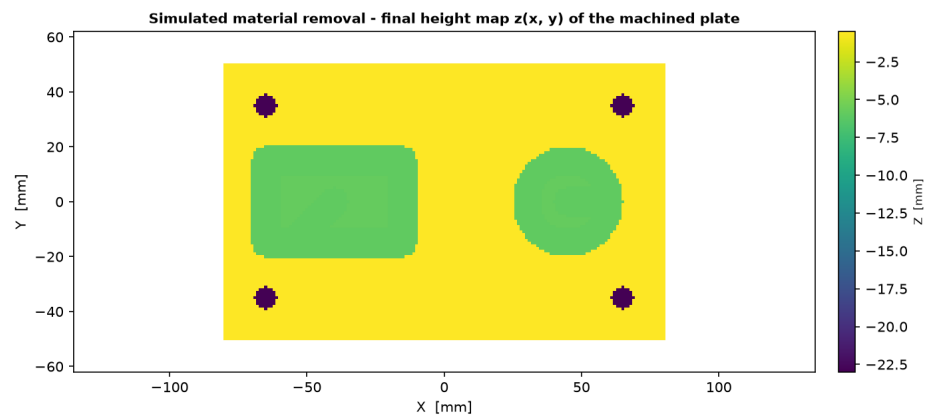
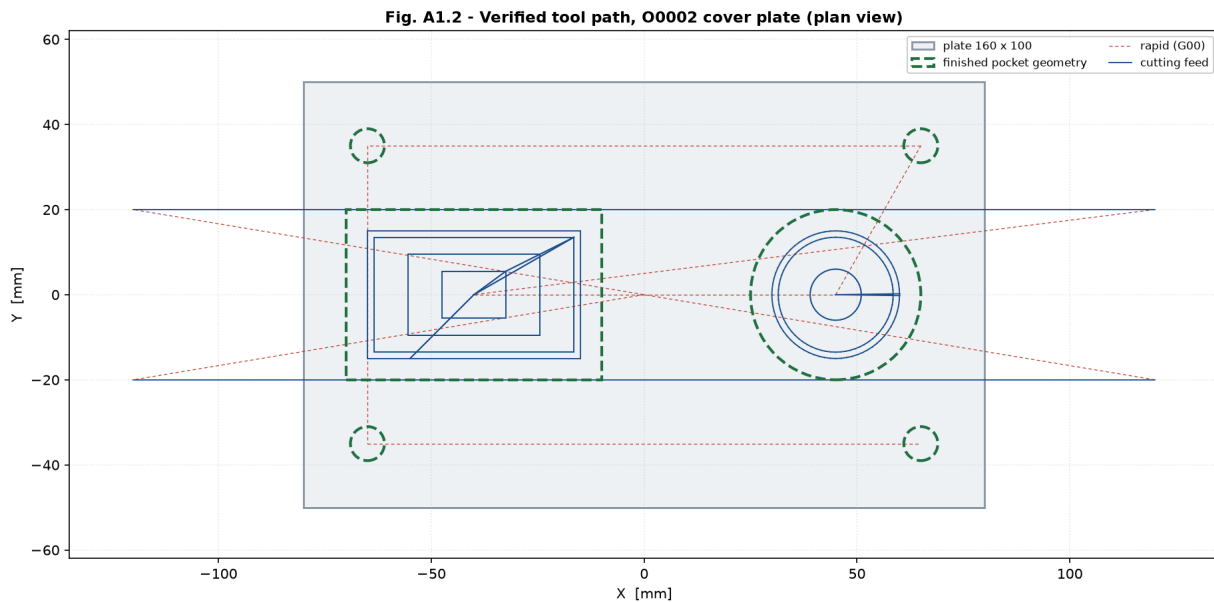


Fig. 2 — Verified tool path, O0002 (upper) and the simulated material-removal height map of the finished plate (lower).

Stated honestly — these are the output of the verifier written for this assignment, **not**

screenshots of CutViewer, LinuxCNC or FreeCAD. The tool is modelled as a cylinder of the declared diameter; holders, turret, fixture and machine kinematics are not modelled, so this does not replace a single-block dry run before the first part is cut.

5. AGV Throughput Calculations

A **unidirectional AGV guide-path loop** serves the cell, outbound (loaded) past the front of every station and returning empty, with five pick-and-deposit stations. Transfer between TC-01 and MC-01 is by a rail-mounted six-axis robot, not by AGV, because that transfer happens every cycle. A roller conveyor was rejected: it fixes the route in steel, whereas the AGV loop can be re-sequenced in software when a new part family is introduced.

$T_c = T_L + L_d/V_c + T_U + L_e/V_c$	delivery cycle time	[min]
$WL = R_f \times T_c$	workload	[vehicle-min per hour]
$AT = 60 \times A \times F_t$	available time	[min per vehicle per hour]
$n = WL / AT$, rounded UP	fleet size	

Input	Symbol	Value	Basis
Loading time	T_L	0.60 min	Transfer plus AGV docking and handshake
Unloading time	T_U	0.50 min	Roll-off transfer at the receiving station
Loaded distance	L_d	140 m	48 m from the central raw store + 28 m through the cell + 64 m to the despatch dock
Empty return distance	L_e	110 m	Direct return leg of the guide path
Vehicle speed	V_c	50 m/min	Safe speed for a laser-guided AGV in a manned aisle
Delivery demand	R_f	22 /h	16 production moves (24 parts/h in pallets of 6, four moves per pallet) + 6 ancillary moves for tooling, empty pallets and swarf
Availability	A	0.95	Opportunity battery charging and breakdowns
Traffic factor	F_t	0.90	Blocking and queueing at the P/D stations

$$\begin{aligned}
 T_c &= 0.60 + 140/50 + 0.50 + 110/50 = 0.60 + 2.80 + 0.50 + 2.20 \\
 &= 6.10 \text{ min per delivery} \\
 WL &= 22 \times 6.10 = 134.20 \text{ vehicle-min per hour} \\
 AT &= 60 \times 0.95 \times 0.90 = 51.30 \text{ min per vehicle per hour} \\
 n &= 134.20 / 51.30 = 2.616 \rightarrow 3 \text{ AGVs} \\
 \text{Fleet utilisation} &= 2.616 / 3 = 87.2 \% \\
 \text{Fleet delivery capacity} &= 3 \times 51.30 / 6.10 = 25.23 \text{ deliveries/h} \\
 \text{Demand} &= 22.00 \text{ deliveries/h} \quad \text{OK}
 \end{aligned}$$

Result	Value
Delivery cycle time, T_c	6.10 min
Workload, WL	134.20 vehicle-min per hour
Available time per vehicle, AT	51.30 min per hour
Vehicles required (exact)	2.616

Result	Value
Fleet size specified	3 AGVs
Fleet utilisation	87.2 %
Fleet delivery capacity	25.23 deliveries/h against a demand of 22.00
Line throughput supported	24 components per hour ($\approx$ 384 per day, two shifts)

Sensitivity. A demand sweep on the same geometry confirms the design is not on a knife edge:

Demand, deliveries/h	5	10	20	30	40	60	80
Vehicles (exact)	0.60	1.19	2.38	3.57	4.76	7.14	9.51
AGVs required	1	2	3	4	5	8	10

Three vehicles cover 15 to 25 deliveries per hour, so there is headroom for a 14 % rise in output before a fourth is needed.

6. Cell Layout Diagram

Fig. A1.3 — Layout of the Automated Flexible Manufacturing Cell (FMC)

plan view · cell envelope 32 m × 18 m · all dimensions in metres

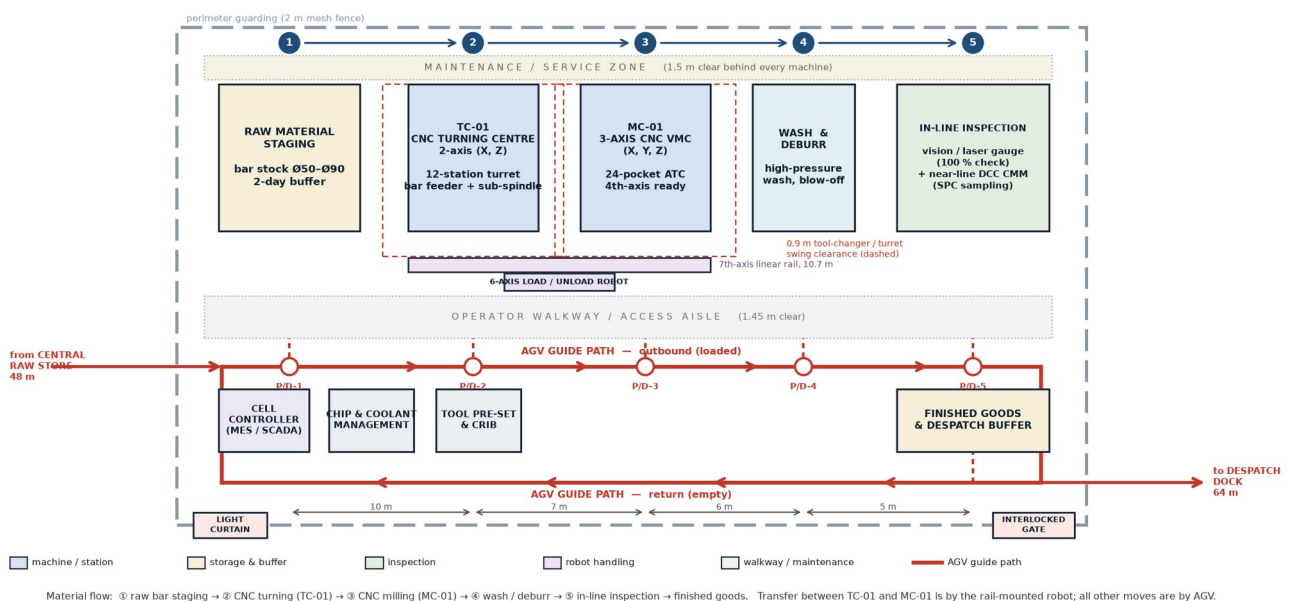


Fig. 3 — Plan layout of the cell, 32 m × 18 m envelope.

- **Machine tool positions** — the five stations sit in one row in process order (raw staging, TC-01, MC-01, wash and deburr, inspection), which keeps the robot rail short and the AGV route a simple one-way loop.
- **Automated material handling path** — a closed AGV loop below the stations, outbound loaded along the machine fronts and returning empty, with five P/D stations. The loop extends outside the cell to the central raw store (48 m) and the despatch dock (64 m).

- **Tool changer clearance** — a 0.9 m clear envelope around TC-01 and MC-01 (shown dashed) for turret indexing and ATC arm swing; neither the robot rail nor the walkway intrudes into it.
- **Operator and maintenance zones** — a 1.45 m operator walkway separates the machine line from the AGV path so nobody stands on the guide path, and a 1.5 m maintenance strip runs behind every machine for access to the cabinet, way lubrication and coolant tank.
- **Safety** — 2 m mesh guarding with one interlocked gate and a light curtain at the operator entry; the AGV loop crosses the guarding line at two openings only.

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O0001 (STEPPED TRANSMISSION SHAFT - MILD STEEL)
(STOCK      : DIA 50 X 100 MM BAR, 1.0 MM FACING ALLOWANCE)
(FINISHED   : DIA 30 X 25 / DIA 40 X 30 / DIA 50 BALANCE)
(FEATURES   : FACING, STEP TURNING, 2X45 CHAMFER, 4X2 GROOVE, M30X1.5 THREAD)
(DATUM      : Z0 = FINISHED FACE, X0 = SPINDLE AXIS, X IS DIAMETER)
(TOOLING)
(T0101 : DCLNR 2525 M12 + CNMG 120408-PR 95 DEG ROUGH / FACE, R0.8)
(T0202 : SDJCR 2525 M11 + DCMT 11T304-PF 93 DEG FINISH, R0.4)
(T0303 : QD-LCH 2525 + QD-NH-0300 3 MM GROOVE / PART-OFF BLADE)
(T0404 : SER 2525 M16 + 16ER AG60 60 DEG EXTERNAL THREADING)
(NOTE : GROOVE BLADE IS REFERENCED ON ITS +Z EDGE)

N10 G21 G40 G97 G99 (METRIC, COMP CANCEL, DIRECT RPM, FEED PER REV)
N20 G28 U0.0 W0.0 (REFERENCE RETURN)

(===== OP 10 : FACING - G94 END FACE CYCLE =====)
N30 T0101
N40 G50 S2500 (CLAMP MAX SPINDLE SPEED FOR CSS)
N50 G96 S180 M03 (CONSTANT SURFACE SPEED 180 M/MIN)
N60 G00 X52.0 Z2.0 M08 (APPROACH, COOLANT ON)
N70 G94 X-1.0 Z0.5 F0.20 (FACING PASS 1 - 0.5 MM)
N80 G94 X-1.0 Z0.0 (FACING PASS 2 - TO FINISHED FACE)
N90 G00 X52.0 Z2.0

(===== OP 20 : ROUGH TURNING - G90 TURNING CYCLE =====)
(LEAVE 0.5 MM ON DIAMETER FOR FINISHING)
N100 G96 S200 M03
N110 G90 X45.0 Z-55.0 F0.30 (ROUGH PASS 1 - DIA 45 OVER 55 MM)
N120 G90 X40.5 Z-55.0 (ROUGH PASS 2 - DIA 40.5, 0.5 STOCK)
N130 G90 X35.5 Z-25.0 (ROUGH PASS 3 - SMALL STEP)
N140 G90 X30.5 Z-25.0 (ROUGH PASS 4 - DIA 30.5, 0.5 STOCK)
N150 G00 X200.0 Z200.0 M09
N160 G28 U0.0 W0.0

(===== OP 30 : FINISH PROFILE - G42 NOSE RADIUS COMP =====)
N170 T0202
N180 G50 S3000
N190 G96 S260 M03
N200 G00 X26.0 Z2.0 M08
N210 G42 G01 Z0.0 F0.12 (NOSE RADIUS COMPENSATION ON)
N220 G01 X30.0 Z-2.0 (2 X 45 DEG CHAMFER)
N230 G01 Z-25.0 (FINISH DIA 30)
N240 G01 X40.0 (SHOULDER FACE TO DIA 40)
N250 G01 Z-55.0 (FINISH DIA 40)
N260 G01 X50.0 (SHOULDER FACE TO DIA 50)
N270 G40 G00 X200.0 Z200.0 M09 (COMPENSATION OFF, RETRACT)
N280 G28 U0.0 W0.0

(===== OP 40 : THREAD RELIEF GROOVE 4 WIDE X 2 DEEP =====)
(GROOVE SPANS Z-20.0 TO Z-24.0, BOTTOM DIA 26.0, 3 MM BLADE, 2 PLUNGES)
N290 T0303
N300 G97 S900 M03 (DIRECT RPM FOR GROOVING)
N310 G00 X32.0 Z-21.0 M08
N320 G01 X26.0 F0.08 (PLUNGE 1 - COVERS Z-24.0 TO Z-21.0)
N330 G04 X0.5 (DWELL TO CLEAN THE GROOVE BOTTOM)
N340 G00 X32.0
N350 G00 Z-20.0
N360 G01 X26.0 F0.08 (PLUNGE 2 - COVERS Z-23.0 TO Z-20.0)
N370 G04 X0.5
N380 G00 X32.0
N390 G00 X200.0 Z200.0 M09
N400 G28 U0.0 W0.0

(===== OP 50 : THREADING M30 X 1.5 - G92 THREAD CYCLE =====)
(MINOR DIA = 30 - 1.2268 X 1.5 = 28.16 MM, DECREASING INFEEED, 1 SPRING PASS)
N410 T0404
N420 G97 S500 M03
N430 G00 X32.0 Z3.0 M08
N440 G92 X29.40 Z-20.0 F1.5 (PASS 1)
N450 X28.96 (PASS 2)
N460 X28.64 (PASS 3)
N470 X28.44 (PASS 4)
N480 X28.30 (PASS 5)
N490 X28.22 (PASS 6)
N500 X28.16 (PASS 7 - FULL DEPTH)
N510 X28.16 (PASS 8 - SPRING PASS)
N520 G00 X200.0 Z200.0 M09
N530 G28 U0.0 W0.0
N540 M05 (SPINDLE STOP)
N550 M30 (END OF PROGRAM, REWIND)
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O0002 (COVER PLATE 160 X 100 X 20 - ALUMINIUM)
(DATUM : X0 Y0 = CENTRE OF THE PLATE, Z0 = FINISHED TOP FACE)
(FEATURES : FACE MILL, 60X40X6 RECT POCKET, DIA 40 X 6 CIRCULAR POCKET,)
( : 4 X DIA 8 THROUGH HOLES ON A 130 X 70 RECTANGULAR PATTERN)
(TOOLING)
(T01 : DIA 63 FACE MILL, 5 INSERTS)
(T02 : DIA 12 3-FLUTE CENTRE-CUTTING SLOT DRILL - POCKET ROUGHING)
(T03 : DIA 10 4-FLUTE END MILL - POCKET FINISHING WITH G41)
(T04 : DIA 8 PARABOLIC FLUTE DRILL - THROUGH HOLES)
(SUBPROGRAMS : 01001 RECT POCKET DEPTH LEVEL, 01002 4-HOLE PATTERN)

N10 G21 G17 G40 G49 G80 G90 (METRIC, XY PLANE, ALL COMP CANCELLED)
N20 G91 G28 Z0.0 (Z TO MACHINE REFERENCE)

(===== OP 10 : FACE MILL THE TOP SURFACE =====)
N30 T01 M06
N40 G90 G54 G00 X-120.0 Y-20.0 S1200 M03
N50 G43 H01 Z50.0 M08 (TOOL LENGTH COMPENSATION ON)
N60 G00 Z-0.5
N70 G01 X120.0 F500 (PASS 1)
N80 G00 Z5.0
N90 G00 X-120.0 Y20.0
N100 G00 Z-0.5
N110 G01 X120.0 F500 (PASS 2)
N120 G00 Z50.0 M09
N130 G91 G28 Z0.0

(===== OP 20 : RECTANGULAR POCKET ROUGHING =====)
(POCKET 60 X 40 X 6 DEEP CENTRED AT X-40.0 Y0.0)
(ROUGH LEAVES 0.5 MM ON THE WALLS AND 0.2 MM ON THE FLOOR)
N140 T02 M06
N150 G90 G54 G00 X-40.0 Y0.0 S2400 M03
N160 G43 H02 Z50.0 M08
N170 G00 Z0.2
N180 M98 P1001 L2 (TWO 3.0 MM DEPTH LEVELS -> Z-5.8)
N190 G90 G00 Z50.0 M09
N200 G91 G28 Z0.0

(===== OP 30 : RECTANGULAR POCKET FINISHING - G41 =====)
N210 T03 M06
N220 G90 G54 G00 X-40.0 Y0.0 S3000 M03
N230 G43 H03 Z50.0 M08
N240 G00 Z1.0
N250 G01 Z-6.0 F120
N260 G01 G41 D03 X-55.0 Y-20.0 F300 (DOWN TO FINISHED DEPTH IN CLEARED STOCK)
N270 G01 X-10.0 Y-20.0 (CUTTER COMP LEFT ON DURING LEAD-IN)
N280 G01 X-10.0 Y20.0 (CONTOUR CORNER 1)
N290 G01 X-70.0 Y20.0 (CONTOUR CORNER 2)
N300 G01 X-70.0 Y-20.0 (CONTOUR CORNER 3)
N310 G01 X-55.0 Y-20.0 (CONTOUR CORNER 4)
N320 G40 G01 X-40.0 Y0.0 (CLOSE THE CONTOUR)
N330 G00 Z50.0 M09 (COMPENSATION OFF, BACK TO CENTRE)
N340 G91 G28 Z0.0

(===== OP 40 : CIRCULAR POCKET DIA 40 X 6 DEEP =====)
(CENTRED AT X45.0 Y0.0 - ROUGHED WITH T02, FINISHED WITH T03 AND G41)
N350 T02 M06
N360 G90 G54 G00 X45.0 Y0.0 S2400 M03
N370 G43 H02 Z50.0 M08
N380 G00 Z1.0
N390 G01 Z-5.8 F150 (PLUNGE AT THE CENTRE, 0.2 FLOOR STOCK)
N400 G01 X51.0 Y0.0 F400 (OUT TO CUTTER-CENTRE RADIUS 6.0)
N410 G03 X51.0 Y0.0 I-6.0 J0.0 (FULL CIRCLE, CCW)
N420 G01 X58.5 Y0.0 (STEP OUT TO RADIUS 13.5)
N430 G03 X58.5 Y0.0 I-13.5 J0.0 (FULL CIRCLE - ROUGH WALL AT DIA 39.0)
N440 G01 X45.0 Y0.0 (BACK TO CENTRE)
N450 G00 Z50.0 M09
N460 G91 G28 Z0.0

N470 T03 M06
N480 G90 G54 G00 X45.0 Y0.0 S3000 M03
N490 G43 H03 Z50.0 M08
N500 G00 Z1.0
N510 G01 Z-6.0 F120
N520 G01 G41 D03 X65.0 Y0.0 F300 (COMP ON, LEAD-IN TO THE CONTOUR)
N530 G03 X65.0 Y0.0 I-20.0 J0.0 (FINISH THE DIA 40 BORE IN ONE CIRCLE)
N540 G40 G01 X45.0 Y0.0 (COMPENSATION OFF)
N550 G00 Z50.0 M09
N560 G91 G28 Z0.0

(===== OP 50 : 4 X DIA 8 THROUGH HOLES - G83 PECK =====)
N570 T04 M06
N580 G90 G54 G00 X65.0 Y35.0 S1800 M03
N590 G43 H04 Z50.0 M08
N600 G00 Z5.0
N610 G99 G83 Z-23.0 R2.0 Q5.0 F150 (PECK CYCLE, 3 MM BREAK-THROUGH)
N620 M98 P1002 (SUBPROGRAM DRILLS THE 4-HOLE PATTERN)
N630 G80 G00 Z50.0 M09 (CANCEL THE CANNED CYCLE)
N640 G91 G28 Z0.0
N650 M05
N660 M30

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%
O1001 (RECT POCKET - ONE DEPTH LEVEL, SPIRAL OUT FROM THE CENTRE)
N10 G91 G01 Z-3.0 F150 (INCREMENTAL STEP DOWN)
N20 G90 G01 X-32.5 Y5.5 F450 (LOOP A)
N30 G01 X-47.5 Y5.5
N40 G01 X-47.5 Y-5.5
N50 G01 X-32.5 Y-5.5
N60 G01 X-32.5 Y5.5
N70 G01 X-24.5 Y9.5 (LOOP B)
N80 G01 X-55.5 Y9.5
N90 G01 X-55.5 Y-9.5
N100 G01 X-24.5 Y-9.5

```

```

N110 G01 X-24.5 Y9.5
N120 G01 X-16.5 Y13.5          (LOOP C - ROUGH WALL AT 0.5 STOCK)
N130 G01 X-63.5 Y13.5
N140 G01 X-63.5 Y-13.5
N150 G01 X-16.5 Y-13.5
N160 G01 X-16.5 Y13.5
N170 G01 X-40.0 Y0.0 F1000     (RETURN TO THE CENTRE IN CLEARED STOCK)
N180 M99
%

%
O1002 (4-HOLE PATTERN - CALLED WITH THE G83 CANNED CYCLE ACTIVE)
N10 G90 X65.0 Y35.0           (HOLE 1)
N20 X-65.0 Y35.0              (HOLE 2)
N30 X-65.0 Y-35.0             (HOLE 3)
N40 X65.0 Y-35.0              (HOLE 4)
N50 M99
%
```

Appendix C — Simulation Verification Log

Execution log : toolpath_simulator.py
Interpreter : Python 3.11.15
Executed on : 2026-10-02 10:48:41 UTC

ASSIGNMENT 1 - G/M-CODE TOOL PATH SIMULATION AND COLLISION VERIFICATION
#####

=====

PROGRAM 00001 - turning_shaft.nc (stepped transmission shaft)

=====

Blocks interpreted into 83 elementary moves
Move breakdown : dwell = 2, feed = 14, rapid = 59, thread = 8
Tools used : T0101, T0202, T0303, T0404
Z range swept : -55.00 to 300.00 mm

VERIFICATION RESULT : PASS
No collisions, no envelope violations and no programming-practice
errors were detected. The tool path is collision free.

=====

tool path plot written to simulation/turning_toolpath.png

=====

PROGRAM 00002 - milling_plate.nc (cover plate)

=====

Blocks interpreted into 381 elementary moves
Move breakdown : feed = 292, rapid = 89
Tools used : T01, T02, T03, T04
Z range swept : -23.00 to 60.00 mm

VERIFICATION RESULT : PASS
No collisions, no envelope violations and no programming-practice
errors were detected. The tool path is collision free.

=====

tool path plot written to simulation/milling_toolpath.png

=====

NEGATIVE CONTROL - deliberately faulty programs that MUST be rejected

=====

Faulty lathe program : 1 collision(s) detected -> CHECKER WORKING
N50: RAPID into material: at Z=-0.23 the tool is at radius 24.53 mm but material remains to radius 25.00 mm
Faulty mill program : 3 fault(s) detected -> CHECKER WORKING
N60: cutting move with no feed rate set
N60: cutting move with the spindle stopped
N50: RAPID into material at X-40.0 Y0.0 Z-1.00: stock still stands at Z=0.00
>>> the verifier correctly rejects known-bad programs, so the PASS
results reported above for 00001 and 00002 are meaningful.

=====

DIMENSIONAL CHECK OF THE SIMULATED PLATE

=====

rectangular pocket floor	at X -40.0 Y 0.0 : Z =	-6.00	(expected -6.00 +/- 0.25)	OK
circular pocket floor	at X 45.0 Y 0.0 : Z =	-6.00	(expected -6.00 +/- 0.25)	OK
faced top surface	at X 0.0 Y 0.0 : Z =	-0.50	(expected -0.50 +/- 0.25)	OK
uncut corner of the plate	at X -78.0 Y -48.0 : Z =	-0.50	(expected -0.50 +/- 0.6)	OK
rect pocket, 2 mm INSIDE +X wall at X -12.0 Y 0.0 : Z =	-6.00	(expected -6.00 +/- 0.25)	OK	
rect pocket, 2 mm OUTSIDE +X wall at X -8.0 Y 0.0 : Z =	-0.50	(expected -0.50 +/- 0.25)	OK	
rect pocket, 2 mm INSIDE +Y wall at X -40.0 Y 18.0 : Z =	-6.00	(expected -6.00 +/- 0.25)	OK	
rect pocket, 2 mm OUTSIDE +Y wall at X -40.0 Y 22.0 : Z =	-0.50	(expected -0.50 +/- 0.25)	OK	
circ pocket, R18 (inside wall) at X 63.0 Y 0.0 : Z =	-6.00	(expected -6.00 +/- 0.25)	OK	
circ pocket, R22 (outside wall) at X 67.0 Y 0.0 : Z =	-0.50	(expected -0.50 +/- 0.25)	OK	
rect pocket, 2 mm OUTSIDE -Y wall at X -55.0 Y -22.0 : Z =	-0.50	(expected -0.50 +/- 0.25)	OK	

=====

OVERALL: ALL PROGRAMS VERIFIED COLLISION FREE

=====

PROCESS EXIT CODE: 0